Numerical Analysis Assignment 2

1. Find all fixed points
(a) x^5 (b) x^2-4x+2 (c) $\frac{8+2x}{2+x^2}$

Solution: (a) $x^5 = x$
$x^5 - x = x(x^4-1) = 0 \Rightarrow x = 0, -1, 1$

(b) $x^2-4x+2 = x$
$x^2-5x+2 = 0 \Rightarrow x = \frac{5 \pm \sqrt{17}}{2}$

(c) $\frac{8+2x}{2+x^2} = x$
$x^3+2x-2x-8 = 0 \Rightarrow x = 2$
$[(x-2)(x^2+2x+4) = 0]$

2. Which of the following ~~the~~ three Fixed-Point Iterations converge to $\sqrt{5}$? Rank the ones that converge from fastest to slowest.
(a) $x \to \frac{4}{5}x + \frac{1}{x}$ (b) $x \to \frac{x}{2} + \frac{5}{2x}$ (c) $x \to \frac{x+5}{x+1}$

Solution: (a) $g(x) = \frac{4}{5}x + \frac{1}{x}$ $g(\sqrt{5}) = \sqrt{5}$ $x = \sqrt{5}$ is a fixed point
$g'(x) = \frac{4}{5} - \frac{1}{x^2}$ $g'(\sqrt{5}) = \frac{3}{5}$ ← linear converge

(b) $g(x) = \frac{x}{2} + \frac{5}{2x}$ $g(\sqrt{5}) = \sqrt{5}$ $x = \sqrt{5}$ is a fixed point
$g'(x) = \frac{1}{2} - \frac{5}{2x^2}$ $g'(\sqrt{5}) = 0$ ← superlinear converge.

(c) $g(x) = \frac{x+5}{x+1}$ $g(\sqrt{5}) = \sqrt{5}$ $x = \sqrt{5}$ is a fixed point
$g'(x) = \frac{-4}{(x+1)^2}$ $g'(\sqrt{5}) = \frac{-4}{(\sqrt{5}+1)^2} \approx -0.3820$ $|g'(\sqrt{5})| = 0.3820$
linear converge.

So (b) is fastest, then (c), (a) is slowest

Fixed-point Iteration.

3. $g(x) = 1 - 5x + \frac{15}{2}x^2 - \frac{5}{2}x^3$

(a) Show $1-\sqrt{3/5}$, 1, $1+\sqrt{3/5}$ are fixed point.

$x = g(x)$

即 $\frac{5}{2}x^3 - \frac{15}{2}x^2 + 6x - 1 = 0$

$x^3 - 3x^2 + \frac{12}{5}x - \frac{2}{5} = 0$

~~$[x-(1-\sqrt{3/5})](x-1)[x-(1+\sqrt{3/5})]$~~

let $x=1$, $1 - 3 + \frac{12}{5} - \frac{2}{5} = 0$ so $x=1$ fixed point.

then $x^3 - 3x^2 + \frac{12}{5}x - \frac{2}{5} = (x-1)(x^2 - 2x + \frac{2}{5}) = 0$

so $x^2 - 2x + \frac{2}{5} = 0$.

$x = \frac{2 \pm \sqrt{4 - 4\cdot\frac{2}{5}}}{2} = 1 \pm \sqrt{\frac{3}{5}}$ fixed points.

(b) Show none of the three fixed points is locally convergent.

$g'(x) = -5 + 15x - \frac{15}{2}x^2$

$|g'(1)| = \frac{5}{2} > 1$

$|g'(1-\sqrt{\frac{3}{5}})| = |-5 + 15(1-\sqrt{\frac{3}{5}}) - \frac{15}{2}(\frac{8}{5} - 2\sqrt{\frac{3}{5}})| = 2 > 1$

$|g'(1+\sqrt{\frac{3}{5}})| = |-5 + 15(1+\sqrt{\frac{3}{5}}) - \frac{15}{2}(\frac{8}{5} + 2\sqrt{\frac{3}{5}})| = 2 > 1$

4. Show that the initial guesses 0, 1, 2 lead to a fixed point in problem 3. What happens to other initial guesses close to those numbers.

Proof: $x_0 = 0$ $\quad x_1 = g(x_0) = g(0) = 1$ ← fixed point.

$x_0 = 1$ $\quad x_1 = g(x_0) = g(1) = 1$ ↙

$x_0 = 2$ $\quad x_1 = g(x_0) = g(2) = 1$ ↙

Because $x_{i+1} - r = g'(c_i)(x_i - r)$

c_i between x_i and r

let $|x_i - r| = e_i$

so $e_{i+1} = |g'(c_i)| e_i$

when x_i close to r,

because $|g'(r)| > 1 \Rightarrow |g'(c_i)| > 1$

so $e_{i+1} > e_i$

x_{i+1} is far from r.